Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

**SDC HACKATHON 2026**

# Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | XMM |
| **Project / Solution Name** | MouManTAI — Intelligent SME Operations & Commitment Assurance |
| **Name (Leader)** | Wong Jia Hui |
| **Email & Phone (Leader)** | wongjiahui.0111@gmail.com \| +60 17-466 2286 |
| **University Affiliation(s)** | Universiti Teknologi Malaysia (UTM) |
| **Live Prototype URL** | https://moumantai-demo-gateway.onrender.com/ |
| **Code Repository URL (public / access granted to judges)** | https://github.com/jiahui-1101/MouManTAI |

# 2. Architecture & Technology Stack

## 2.1 System Architecture Overview

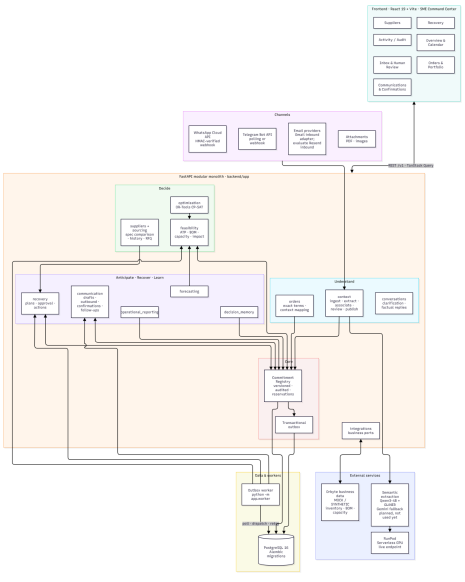
*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

Customers and suppliers write to the SME through Telegram, WhatsApp or Email, using adapters we built for each channel. The FastAPI backend saves every message first, then reads it in layers. Simple rules go first. If they cannot settle the meaning, Qwen3 and GLiNER will run on a RunPod Serverless GPU to help pull out the order details. PyMuPDF and PaddleOCR read PDFs and images. Anything a model says is checked against the original text and unclear cases go to a person.

Clear orders wait for the SME to accept. A feasibility engine checks stock, expiry dates, recipes and capacity. When orders compete for the same resources, Google OR-Tools CP-SAT compares three plans, and the feasibility engine checks each plan again. When a supplier is late, the system finds the affected orders and suggests recovery options. Nothing runs until a manager approves it. The React frontend is the SME Command Center. PostgreSQL stores commitments, approvals and the audit trail, and a background worker passes events between modules. Stock, recipe and capacity data come from a mock Orbyte adapter.

**Block Diagram:**
https://drive.google.com/file/d/1wZLCH7w1sx4F3cqQ6S19wTS_NsslhE0i/view?usp=sharing



## 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| Python 3.11 to 3.14, FastAPI, Uvicorn, Pydantic v2 | Backend API and data validation | PSF / MIT / BSD-3-Clause / MIT |
| SQLAlchemy 2, Alembic, psycopg 3 | Database access and migrations | MIT / MIT / LGPL-3.0 |
| PostgreSQL 16 | Main database | PostgreSQL License |
| React 19, TypeScript 5.9, Vite 7, React Router 7, TanStack Query 5 | SME Command Center frontend | MIT / Apache-2.0 / MIT |
| Vitest 5, Testing Library, openapi-typescript | Frontend tests and API types | MIT |
| Google OR-Tools 9 (CP-SAT) | Multi-order planning (FCFS, Profit, Expiry-Aware) | Apache-2.0 |
| PyMuPDF, PaddleOCR | Read PDFs and images | AGPL-3.0 / Apache-2.0 |
| GLiNER, Transformers, PyTorch, RunPod SDK | Model worker on the GPU endpoint | Apache-2.0 / Apache-2.0 / BSD-3-Clause / MIT |
| httpx, pytest, Docker Compose | Channel clients, backend tests, local run | BSD-3-Clause / MIT / Apache-2.0 |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Telegram Bot API | Receive and send customer and supplier messages | Free |
| WhatsApp Cloud API (Meta, test number) | Receive and send customer and supplier messages | Free test tier |
| Gmail API (Google, OAuth) | Read incoming email | Free |
| RunPod Serverless endpoint | Run Qwen3 and GLiNER to read messy messages. The endpoint is live. | Sponsor-provided (RunPod credits) |
| Mock Orbyte adapter | Stock, recipe and capacity data. Mock data made by the team. | Free (team-built) |
| Gemini API (Google) | Planned backup when Qwen fails. Not used yet. | Not used |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit Workspace | Team development only. Not used for hosting. | Sponsor credits |
| RunPod Serverless GPU | Qwen3 and GLiNER inference (live endpoint) | Sponsor credits |
| GitHub and GitHub Actions | Source control and automatic tests | Free |
| Docker Compose (local machine) | Runs database, API and worker | Free, local |

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Vercel & Render | Deploying and hosting our web solution | Free (Hobby plan ) |

# 4. AI Tools, Models & Datasets

## 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Qwen3-4B-Instruct-2507 (rev. cdbee75) | Alibaba Qwen / Hugging Face; served on RunPod | Pre-trained; message interpretation and field extraction | Apache-2.0 |
| GLiNER multi v2.1 (urchade/gliner_multi-v2.1, rev. 443d26d) | Hugging Face; served on RunPod | Pre-trained; multilingual entity extraction | Apache-2.0 |
| PaddleOCR (PP-OCR) | PaddlePaddle | Pre-trained; image-to-text for attachments | Apache-2.0 |
| Naive, Seasonal Naive, Moving Average | Written by the team | Simple statistics, no training | Team-owned |

## 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Demo orders, messages and commitments | Team-made, in the repository (seed script and fixtures) | Synthetic | Team-owned | N |
| Product name list in several languages (catalog.json) | Team-made, in the repository | Self-made | Team-owned | N |
| Mock Orbyte stock, recipe, capacity and supplier data | Team-made, in the repository | Synthetic | Team-owned | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Demand history and past cases for Forecasting and Decision Memory | Team-made, labelled DEMO in the app | Synthetic | Team-owned | N |
| Real customer or SME records | Not used | None | N/A | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| ChatGPT (OpenAI) | Requirements analysis and document drafting | Major |
| OpenAI Codex | Code generation and code review | Major |
| Replit AI (chat) | Final editing of submission documents | Moderate |
| Claude (Anthropic) | Drafting submission documents | Moderate |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Every change goes through a GitHub pull request and is reviewed by other team members before merging. CI runs backend pytest including a PostgreSQL run that fails if any test is skipped, plus frontend type checks, Vitest and a secret scan. We also wrote tests for the risky rules such as an order cannot be accepted after its terms change and that a recovery plan cannot run without manager approval. Documents written with AI are compared against the running system. The team takes responsibility for everything we submit.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit (official SDC offer) | USD 25 credits per member — Deployments, PostgreSQL, AI Integrations, Agent | Help in planning implementation, resolving error |
| RunPod (official SDC offer) | USD 22.56 GPU credits — Serverless endpoint | Qwen3 and GLiNER inference for message interpretation |
| SDC workshops | GenAI, Database and Cloud Deployment workshops | Architecture, database design and deployment approach |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Yes. RunPod is a real part of our system. Our backend calls a live RunPod Serverless endpoint that runs Qwen3 and GLiNER, so MouManTAI can read messy orders and supplier messages in English, Malay and Chinese. The SDC Cloud Deployment and GenAI workshops showed us how to set up and call a serverless GPU endpoint, and we built it into the message flow. Without it, the system still works, but more messages go to Human Review. Feasibility, planning and approvals run in our own backend, so a provider change does not break business rules.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

We only use synthetic data made by the team, so no real customer data is collected. The system stores customer and supplier names and contact IDs, messages, attachments, order terms, user roles, approvals and audit logs. Every record belongs to one tenant, and the server checks this on every request, not only on the screen. Message text is sent to the RunPod model endpoint only when rules cannot read it, and only the text needed is sent. Customer replies are built from approved templates on our own server, so no private data goes to any other AI service. Secrets stay in environment variables, the repository only has placeholders, and a secret scan runs in CI. The current login uses test headers, so it is for local use only. A real deployment would need proper login, consent, and a clear retention rule under Malaysia's PDPA 2010.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

Language models can misread products, dates or intent, so we treat their output as a suggestion. Rules and a source-text check validate every field, and unclear cases go to Human Review. Qwen3-4B is small, so messages that mix Malay, English and Chinese may need more human review. We never trust the solver directly, because the feasibility engine checks every OR-Tools plan again. The profit plan is switched off when price or cost data is missing. Forecasts are advice only and cannot trigger any action. Supplier claims are kept apart from verified delivery history. Stock, recipe and capacity data are mock data, and live Orbyte is not connected. External services can be slow or fail, so we use templates, safe retries and manual review as backups. Emergency actions always need manager approval and a fresh check before they run. Real channel testing is still limited.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

WONG — WONG JIA HUI, TEAM LEADER

25 October 2026